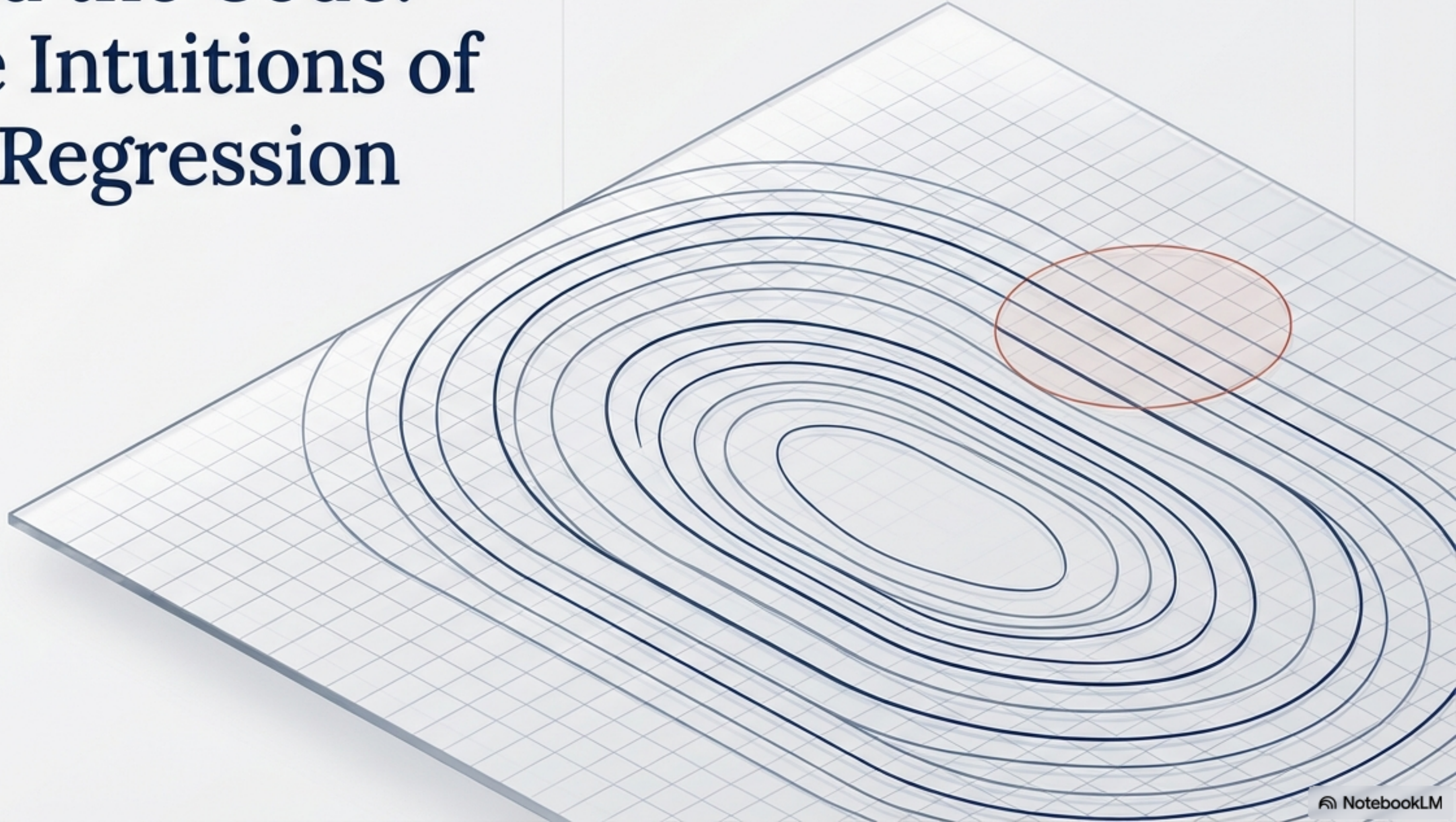


Beyond the Code: 5 Core Intuitions of Ridge Regression



Question: What mathematically happens when we regularise a model?

Answer: We penalise complexity by adding a constraint to the loss function.

$$\text{Original Loss (MSE)} + \text{Penalty Term } (\lambda \sum w^2) = \text{Regularised Loss Function}$$

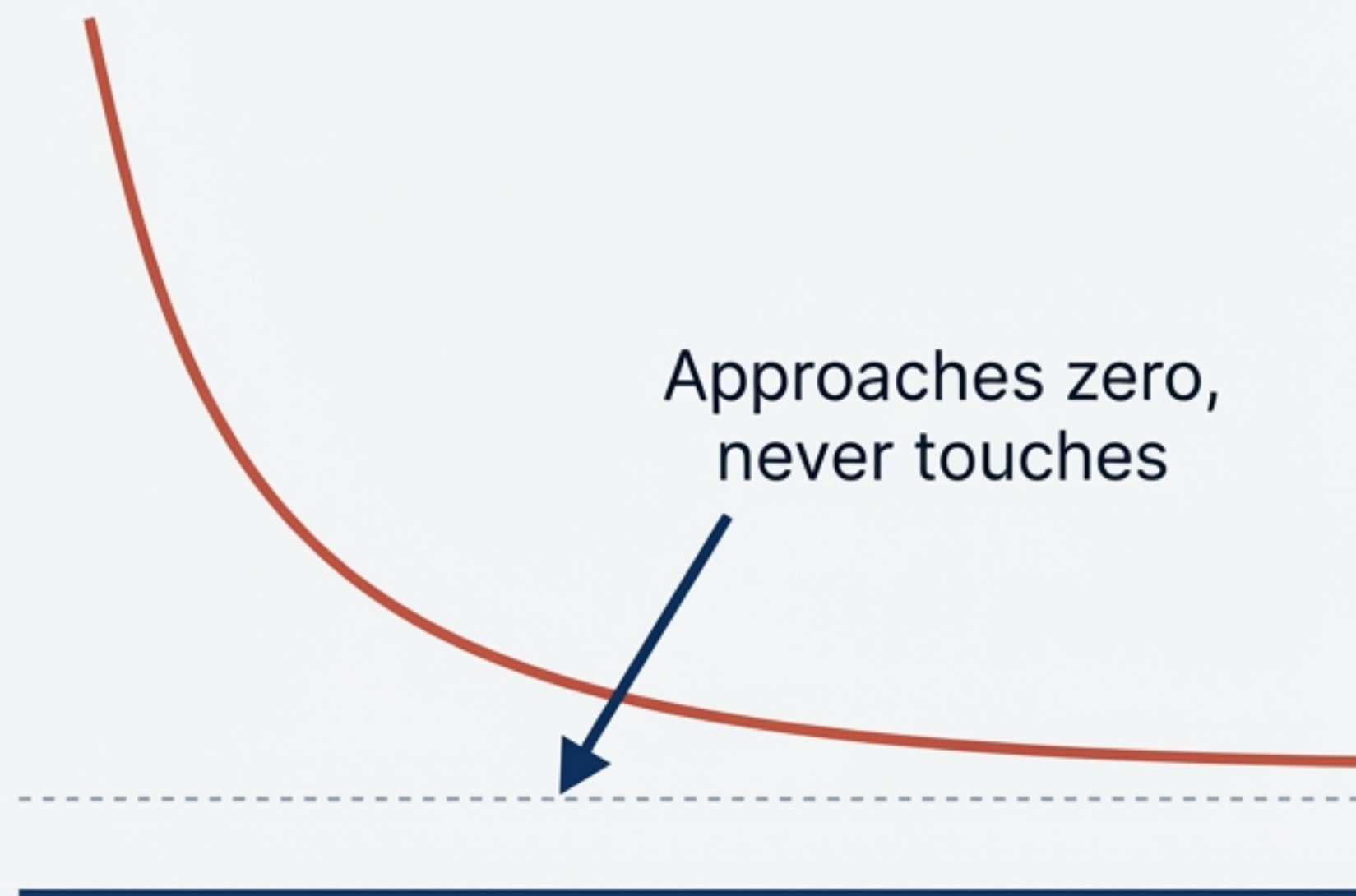


The engine of Ridge Regression

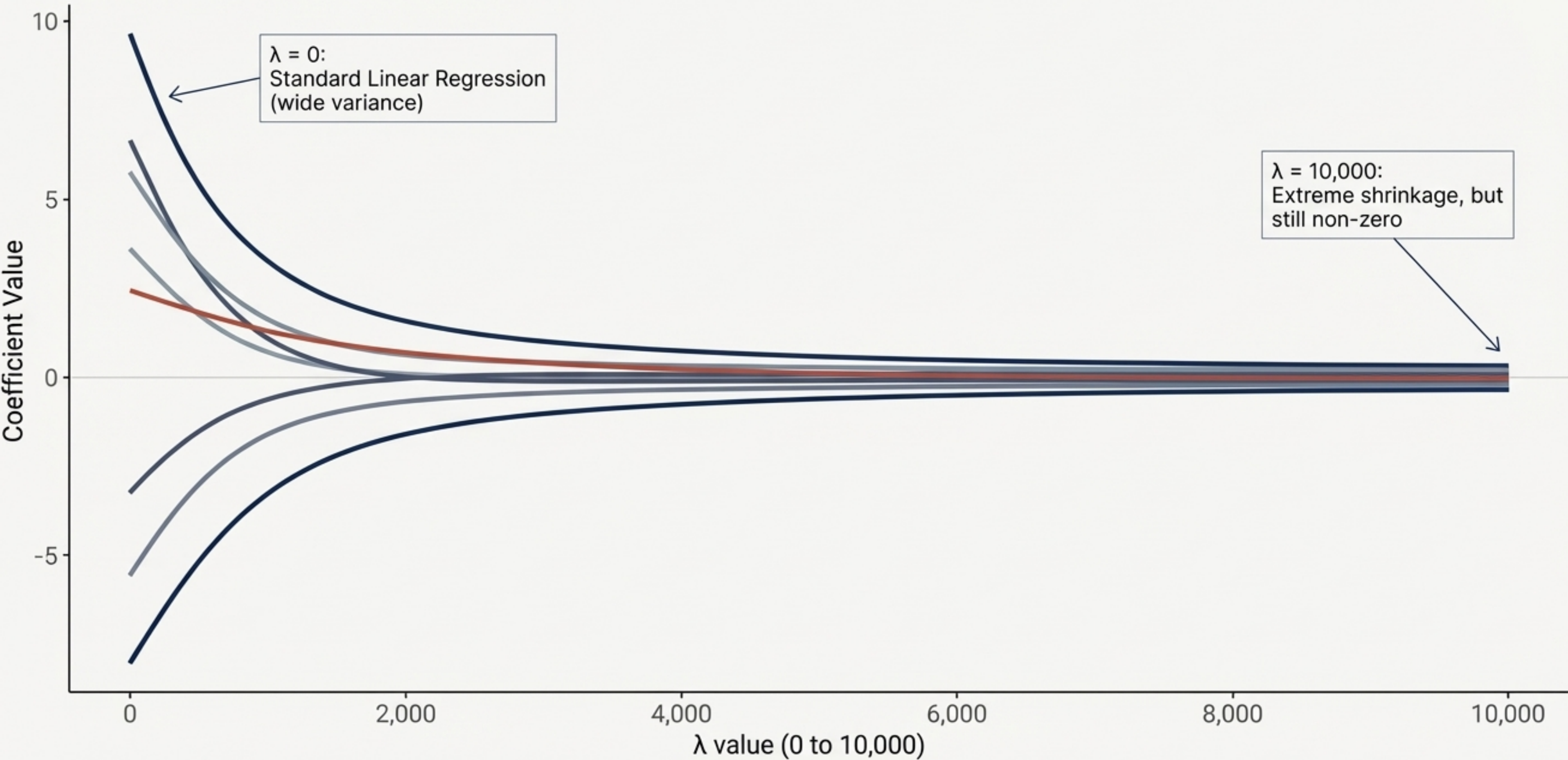
Intuition 1: The Shrinkage Effect

Question: What happens to coefficients as λ increases?

Answer: They shrink progressively towards zero, but never reach absolute zero.



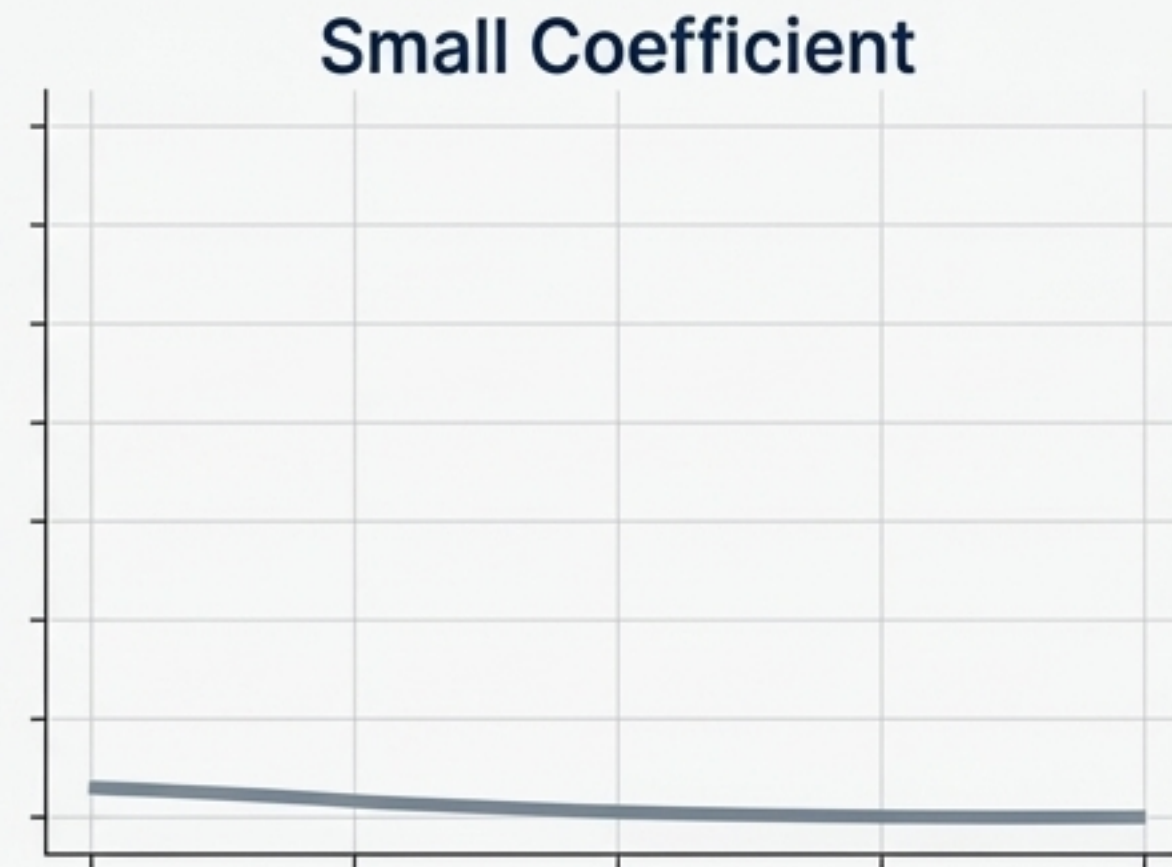
Visualising Shrinkage Across λ



Intuition 2: The Disproportionate Impact

Question: Do all coefficients shrink at the exact same rate?

Answer: No. Larger coefficients experience aggressive shrinkage, while smaller coefficients decrease much more gradually.



The penalty term ($\sum w^2$) squares the weights, punishing large outliers heavily.

Intuition 3: The Bias-Variance Trade-off

Question: How does regularisation impact model generalisation?

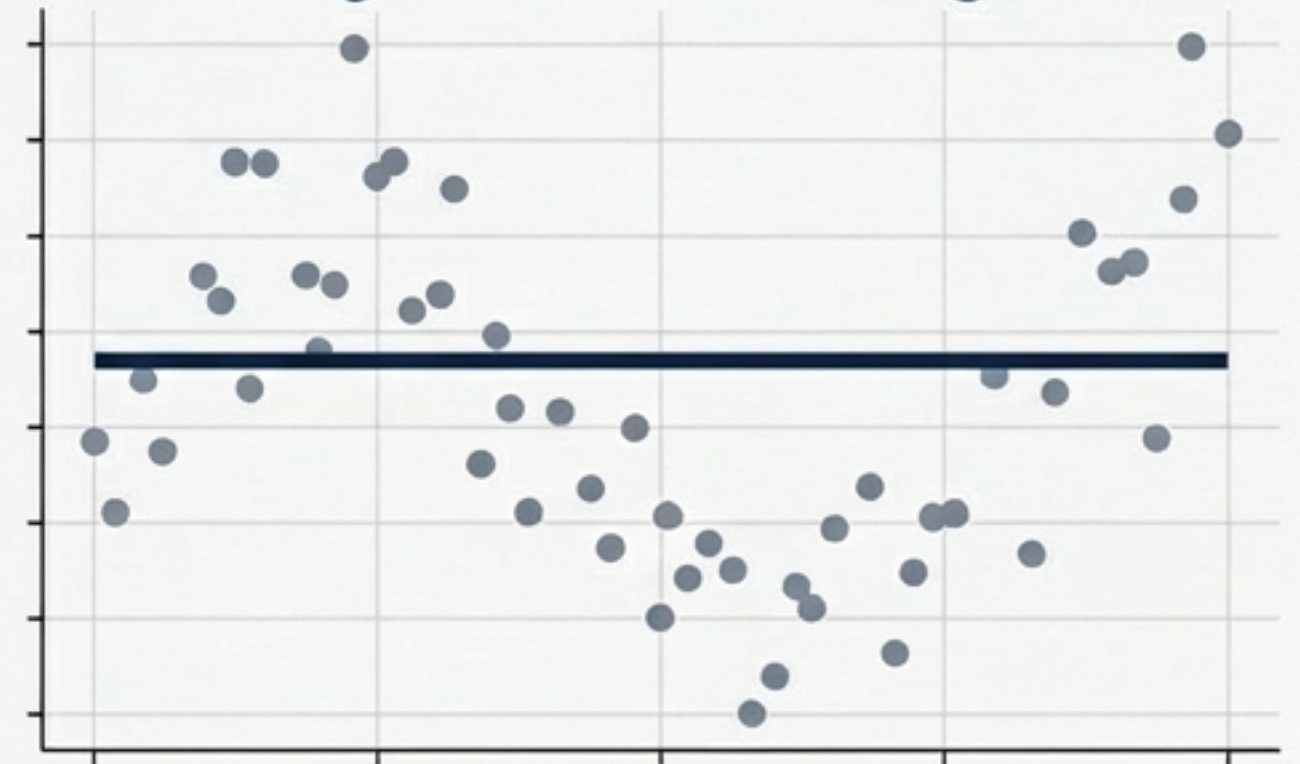
Answer: It forces a mathematical compromise.

- Low λ = Over-fitting (Low Bias, High Variance).
- High λ = Under-fitting (High Bias, Low Variance).

Low λ (Over-fitting)



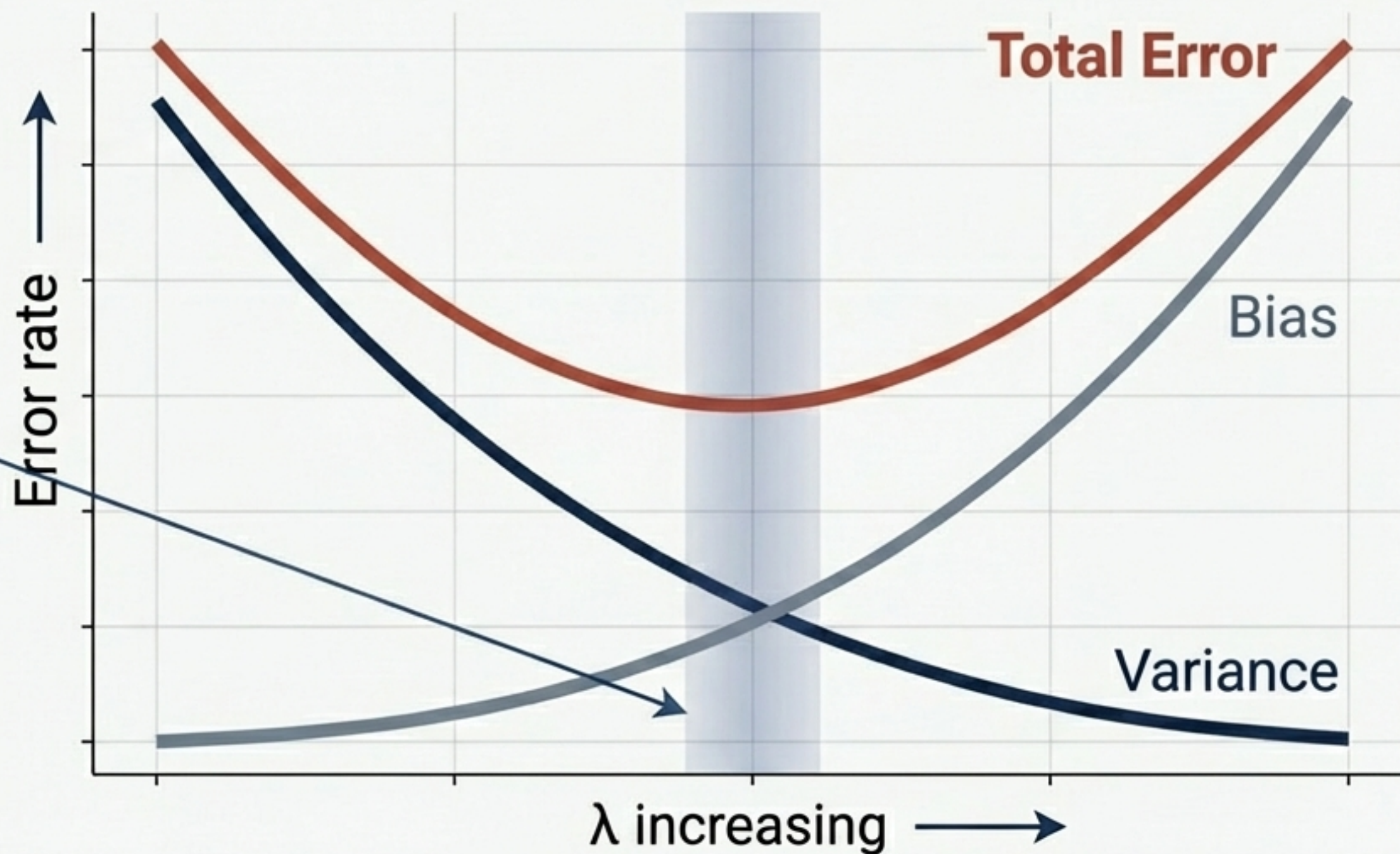
High λ (Under-fitting)



Finding the Sweet Spot

Intuition 4: The Bias-Variance Trade-off

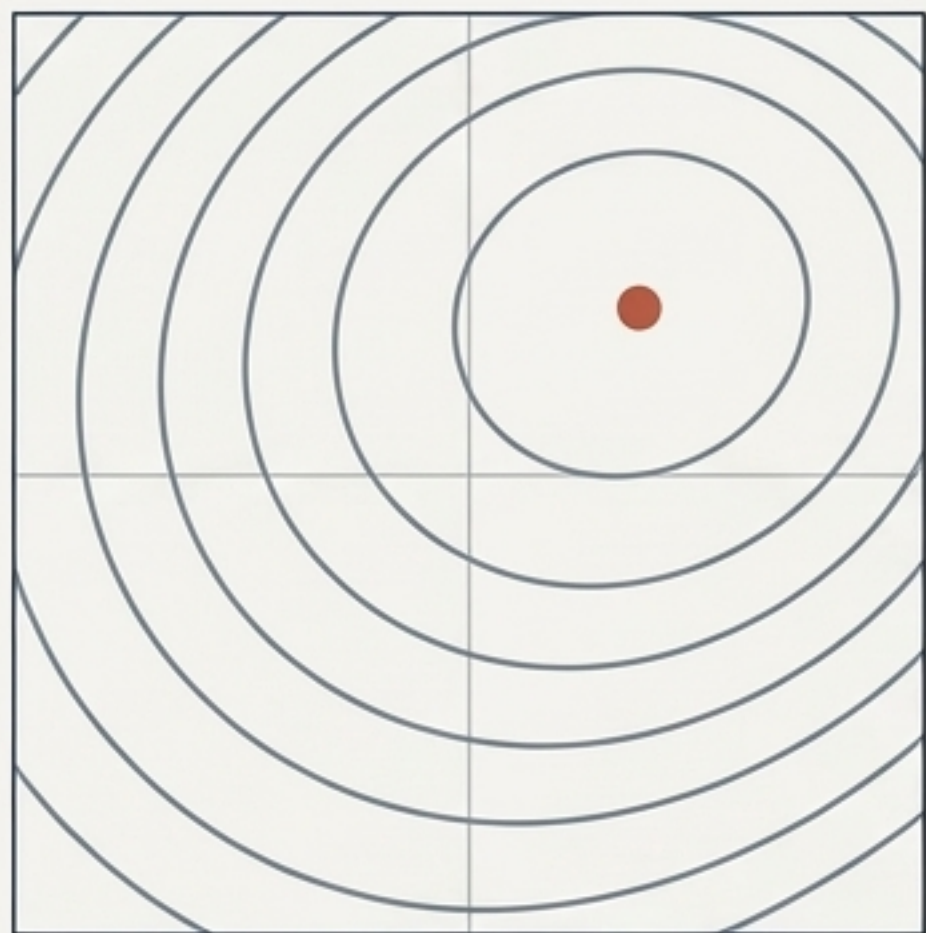
The optimal λ minimises total generalisation error.



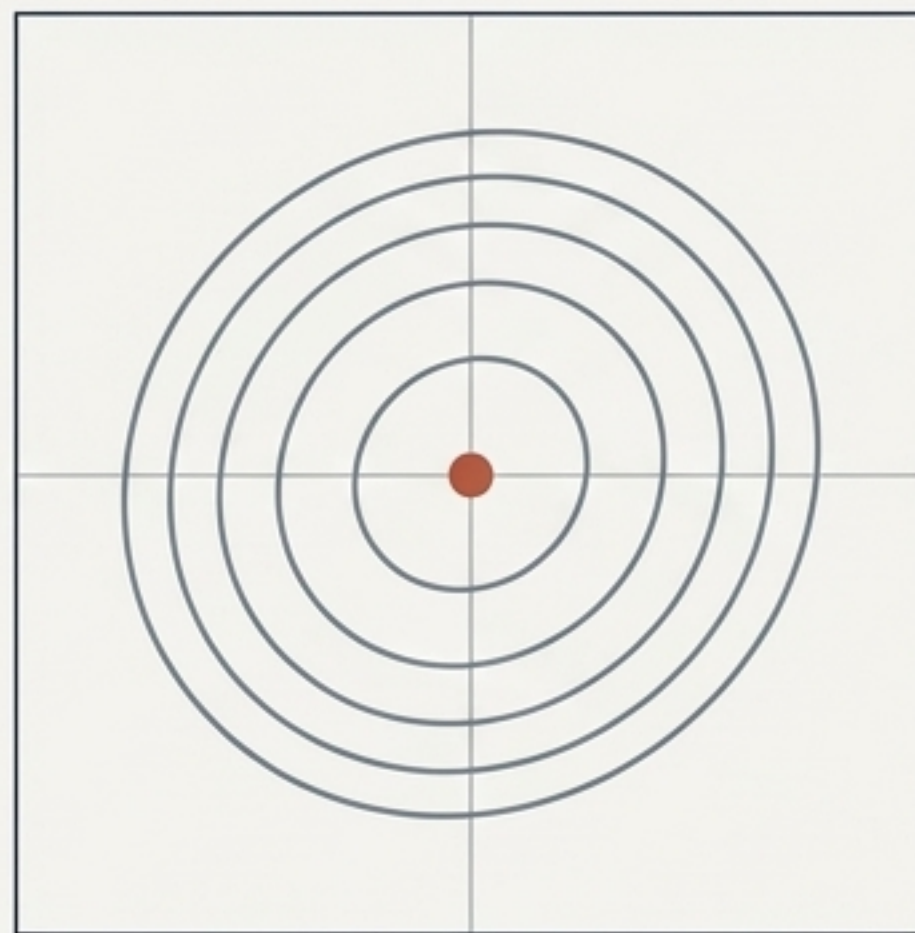
Intuition 4: The Shifting Loss Function

Question: What happens to the geometry of the loss function?

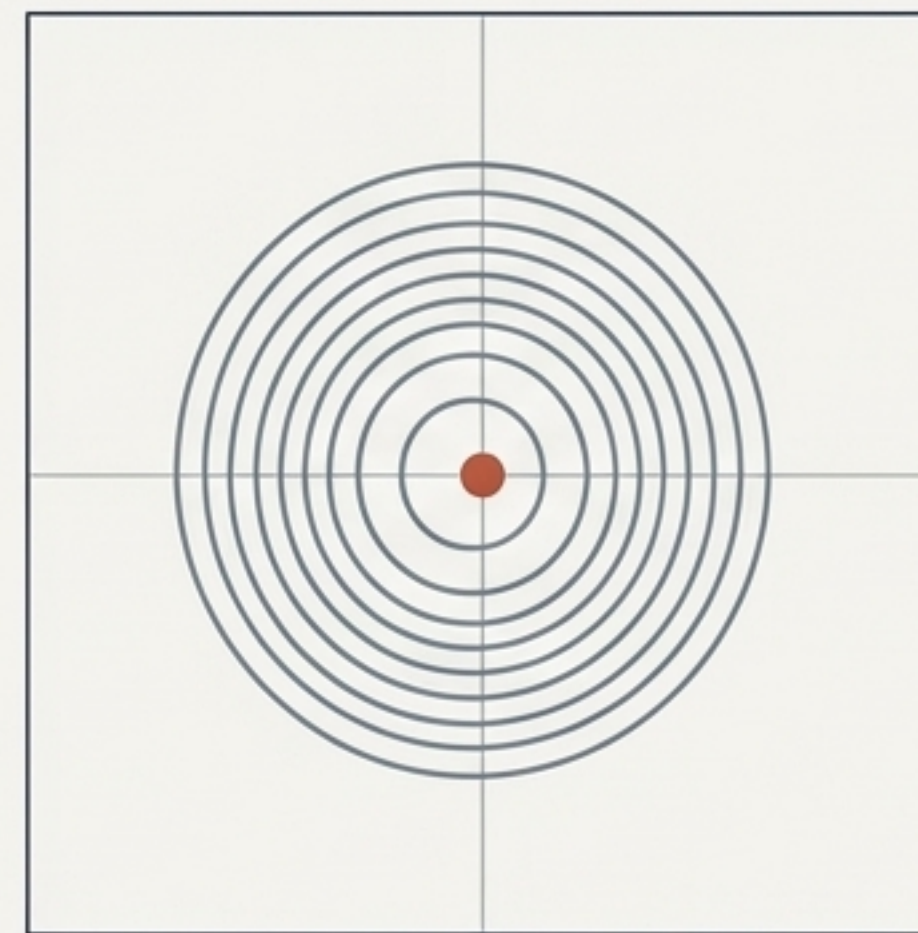
Answer: As λ increases, the global minimum of the loss function is physically dragged towards the origin (0,0).



Plot A ($\lambda = 0$)



Plot B (Moderate λ)



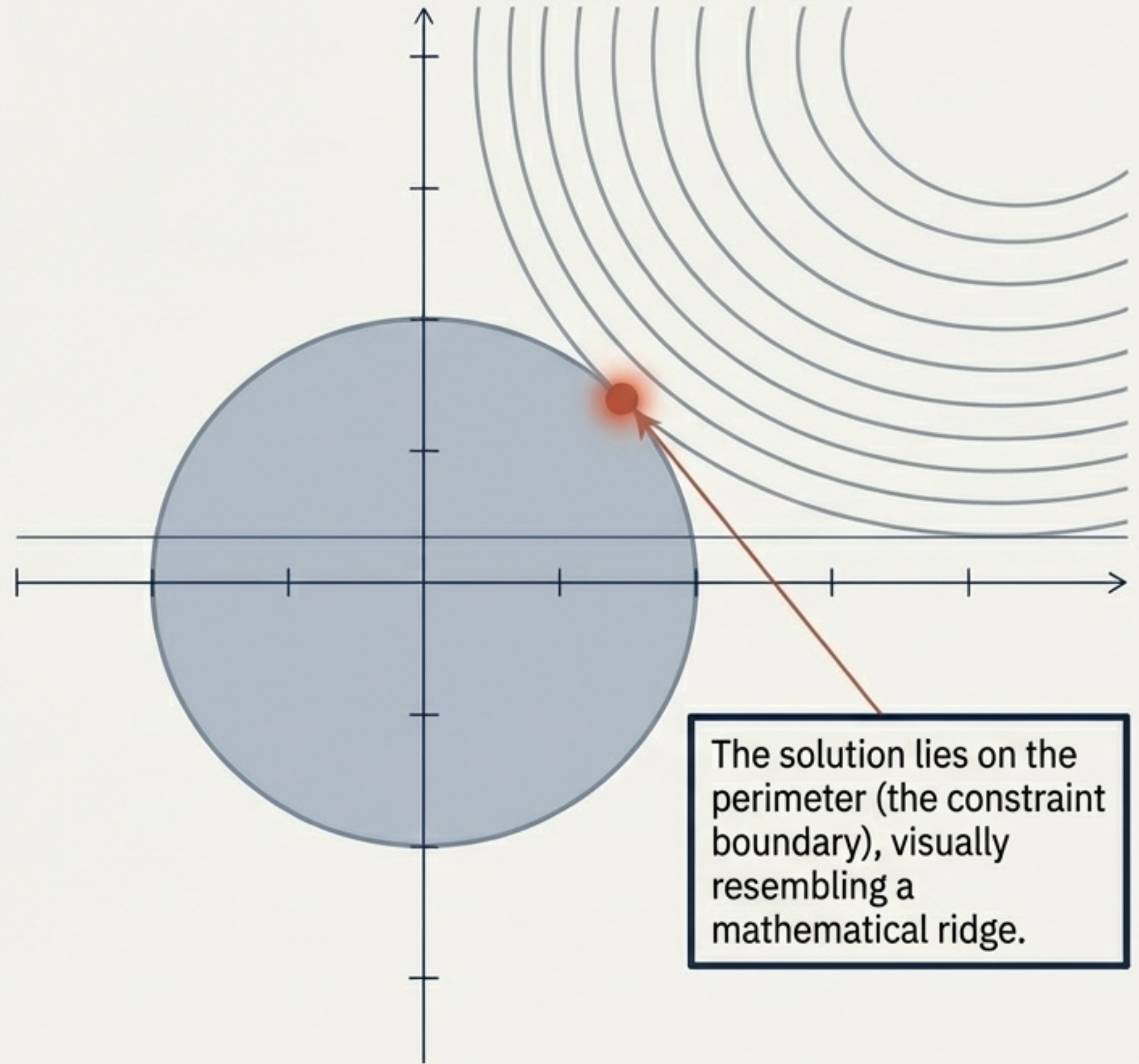
Plot C (High λ)

Intuition 5: Why is it called “Ridge”?

Question: Where does the terminology originate?

Answer: The optimal solution is forced onto the boundary of a constraint region.

The solution lies on the perimeter (the constraint boundary), visually resembling a mathematical ridge.

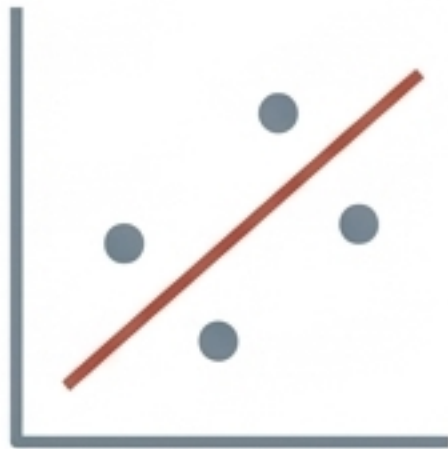


The Golden Rule of Application

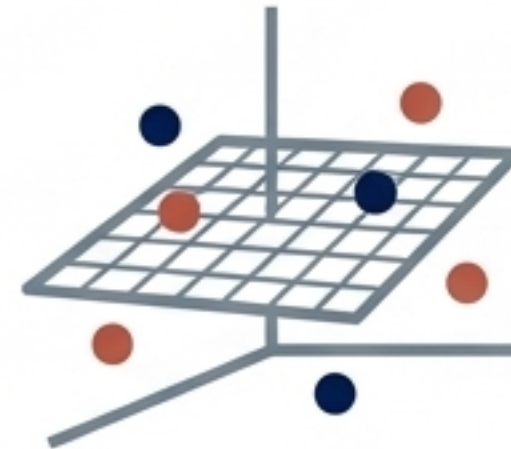
Question: When should you actively use Ridge Regression?

Answer: Only when dealing with multiple input features ($N \geq 2$).

1 Feature $\rightarrow$ Standard Linear Regression



2+ Features $\rightarrow$ Ridge Regression



Regularisation is redundant for a single variable; there is no multi-variable over-fitting to penalise.

Interview Cheat Sheet: Ridge Regression

Concept	Mathematical Mechanism	Practical Takeaway
Shrinkage	$\lambda \uparrow$ compresses w	$w \rightarrow 0$, but never exactly 0.
Impact	Penalty squares weights	Large coefficients shrink faster.
Trade-off	Shifts fit complexity	Balances Bias and Variance.
Geometry	Alters loss function shape	Global minimum shifts to origin.
The "Ridge"	Introduces hard constraints	Solution sits on the constraint boundary.

True mastery of machine learning
lies not in importing libraries,
but in understanding the
geometric and mathematical
consequences of your code.